POSTER · UNIT IV

Parts of a quadcopter

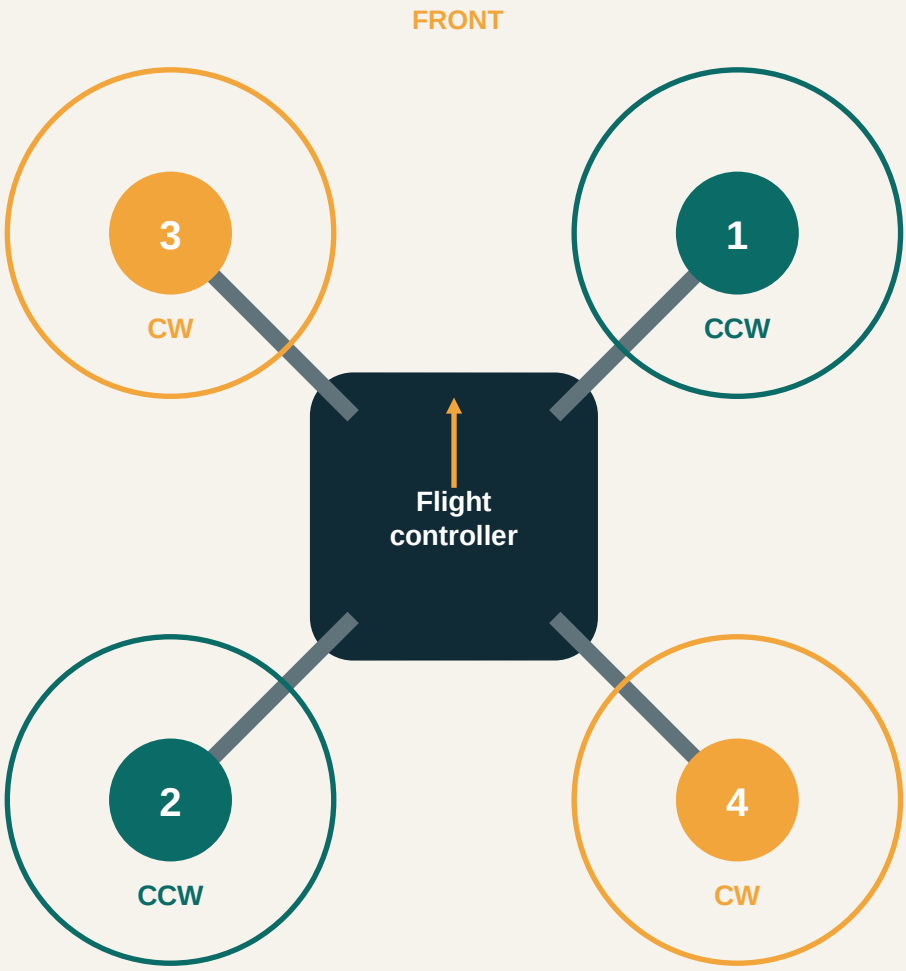
Quad-X, top view. Adjacent motors spin in opposite directions

Propeller

Diameter × pitch (e.g. 10×4.5). CW and CCW props are different parts.

BLDC motor

KV = rpm per volt, no load. Order and direction checked with props OFF.



ESC

Electronic speed controller. PWM needs calibration; DShot does not.

Battery (LiPo)

S count sets voltage, Ah sets capacity, C rating sets max current.

PDB / BEC

Splits battery power; BEC gives clean 5 V to the avionics.

GNSS + compass

On a mast, away from power wires and motors.

RC receiver / telemetry radio

C2 uplink and telemetry downlink to the GCS.

Frame

Arms + center plates; soft-mount the FC against vibration.

Motor numbers follow ArduPilot Quad-X; PX4 numbers differ. Check your firmware's diagram.